

Inverse-Designed Suspended Silicon-Nitride Photonic-Crystal Cavities for Atom-Photon Transduction at the Rubidium D2 Line

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Scaling neutral-atom quantum processors into modular networks requires deterministic, high-efficiency interfaces that map a trapped atom’s state onto a propagating photon. We present a suspended silicon-nitride one-dimensional photonic-crystal nanobeam cavity, inverse-designed for the ^{87}Rb D2 line (780 nm), that directly maximizes the atom-to-waveguide transduction efficiency rather than the bare quality factor. Using differentiable adjoint finite-difference time-domain simulation, we co-optimize per-hole lattice constants, hole sizes, and free-form closed spline hole shapes, seeded by a Bloch band-engineered mirror cell; a single conformal in-plane scale then tunes the resonance onto the atomic line. The absolute resonance is pinned to the D2 line to within one cavity linewidth by a grid-offset-averaged, mesh-converged frequency analysis. From the classical electromagnetic fields we extract a chip-level interface efficiency $\eta_{\text{chip}} = \beta C/(C+1) = 0.84$ (or $\eta = 0.83$ with an assumed $\eta_{\text{fib}} = 0.99$ waveguide-to-fiber factor), with a guided-mode waveguide branching ratio $\beta = 0.88$ validated by mode expansion (99.5% fundamental-mode purity), cooperativity $C \approx 22$, single-photon coupling $g/2\pi \approx 1.0$ GHz at a realistic 250 nm trap height, and mode volume $0.89 (\lambda/n)^3$ — an order-of-magnitude gain in η over a hand-tuned baseline. The design is robust across a fabrication corner grid. All atom-photon figures of merit are fixed-position, two-level, dipole-aligned scalar proxies derived from classical fields, not outputs of a cavity-QED simulation.

I. INTRODUCTION

Arrays of individually trapped neutral atoms have emerged as a leading platform for quantum computation and simulation. Optical tweezers assemble defect-free registers of hundreds of identical qubits [1, 2], whose states are stored in long-lived hyperfine or clock transitions and entangled through strong, switchable Rydberg interactions with gate fidelities now exceeding 99% [3]. Coherent transport of trapped atoms enables reconfigurable, nonlocal connectivity within a single processor [4]. Relative to trapped-ion processors—where a shared motional bus couples qubits but ultimately limits the size of a single crystal [5]—neutral-atom arrays offer a more favorable route to large qubit numbers and dynamically rewirable connectivity, at the cost of weaker native long-range coupling between distant registers.

Bridging that gap, and scaling beyond a single processor, calls for a *photonic* interconnect: a quantum channel that converts a stationary atomic qubit into a flying optical photon and back [6]. Cavity quantum electrodynamics provides the enabling mechanism. An optical resonator enhances the atom’s emission into a well-defined mode and routes it into a waveguide, so that two remote nodes can be heralded into an entangled state by interfering their emitted photons [7, 8]. Integrated nanophotonic implementations are especially attractive for atom arrays because they are compact, support many parallel channels on one chip, and can be aligned to the atomic geometry [9, 10].

The rate and fidelity of such an interconnect are governed by a single figure of merit: the efficiency η with

which an intracavity excitation is delivered into the collection fiber. Two-node entanglement rates scale as η^2 , and the residual infidelity of cavity-mediated protocols falls only as $\ln C/C$ with the cooperativity C [7, 11]. Reaching high η simultaneously demands (i) a large Purcell enhancement, so the atom preferentially emits into the cavity; (ii) strong *over-coupling*, so the cavity decays predominantly into the collection waveguide rather than into scattering or absorption; and (iii) precise resonance with the atomic transition. The first photonic-crystal-waveguide cavity coupled to a single ^{87}Rb atom reached a cooperativity $C \equiv 4g^2/\kappa\gamma \approx 0.07$ [12], so only a few percent of the emission, $C/(C+1)$, was funneled into the cavity mode; later nanophotonic atom-cavity systems have since reached cooperativities of tens, and subsequent integrated nodes identified the waveguide out-coupling fraction as a dominant loss [10, 11, 13, 14]. Closing this gap — raising both the cooperativity and the guided out-coupling — is the key photonic bottleneck to scalable atom-array interconnects.

We target a specific architecture for that interconnect, sketched in Fig. 1(a): an array of suspended photonic-crystal “fingers” cantilevered beyond an etched substrate edge. Atoms held in optical tweezers are transported over the series of cavities, and at each finger the atomic state is transduced into a flying photonic qubit that is collected into fiber. Realizing this picture at scale requires a cavity that is simultaneously low-mode-volume, strongly over-coupled, fabrication-robust, and *exactly* resonant with the atomic line. Silicon nitride is a natural host: it is transparent and low-loss at 780 nm, and a released nanobeam exposes the cavity field to the vacuum mode where the atom is trapped. The design space—hole spacing, hole size, and hole shape, varied independently along the beam—is high-dimensional and nonintuitive, which

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makes it well suited to adjoint-based inverse design rather than hand tuning [15, 16], building on the deterministic gentle-confinement principles developed for ultra-high- Q nanobeams [17–19].

In this work we design such a cavity. Starting from a Bloch band-engineered seed whose mirror gap is centered on the ^{87}Rb D2 line, we use differentiable finite-difference time-domain (FDTD) simulation to co-optimize the per-hole lattice constant, the hole size, and a free-form closed spline hole shape, with a loss function built directly on the transduction efficiency η rather than on Q or C alone. A final conformal in-plane scale tunes the resonance onto the atomic line, which a grid-offset-averaged mesh-convergence analysis pins to within one cavity linewidth. The resulting device reaches a guided-mode branching ratio $\beta = 0.88$ and a chip efficiency $\eta_{\text{chip}} = 0.84$ —a tenfold improvement over a hand-tuned baseline—with a cooperativity $C \approx 22$, two orders of magnitude above the first single-atom photonic-crystal-cavity demonstration ($C \approx 0.07$ [12]) and comparable to the best recent nanophotonic atom-cavity systems, while remaining single-mode and robust across a fabrication corner grid. We emphasize at the outset that all atom–photon figures of merit reported here are fixed-position, two-level, dipole-aligned scalar proxies extracted from classical electromagnetic fields; we make no claim about gate, read-out, or entanglement fidelities, atom motion, or multi-level dynamics, which would require a full cavity-QED treatment beyond the present scope.

II. METHODS

A. Physical platform

The device is a suspended SiN nanobeam of thickness 150 nm and width 500 nm; both are fixed fabrication parameters set by the deposited film and the lithography, and the thickness places the first guided band edge so that the mirror gap can bracket 384 THz. A one-dimensional array of etched holes forms a photonic-crystal mirror on either side of a central defect. The cavity is deliberately asymmetric: a short coupling mirror on the output ($-x$) side and a long reflector on the back ($+x$) side, so that the intracavity field leaks preferentially into a single output waveguide. The atom is trapped in the retro-reflected tweezer antinode 200–300 nm above the top surface, above the Casimir–Polder–limited approach distance; we evaluate all coupling quantities at the nominal 250 nm height.

B. Figures of merit

All atom–photon quantities below are computed as scalar proxies from the simulated classical fields; they are not outputs of a master-equation or quantum-trajectory calculation. The single-photon coupling rate between an

atom of dipole moment μ and the cavity vacuum field is

$$g = \mu \sqrt{\frac{\omega_c}{2\hbar\epsilon_0 V}}, \quad (1)$$

with V the mode volume. Equivalently, sampling the simulated mode at the atom position $\mathbf{r}$,

$$g(\mathbf{r}) = \mu \sqrt{\frac{\omega_c u(\mathbf{r})}{2\hbar\epsilon_0 \int \epsilon |\mathbf{E}|^2 dV}}, \quad u(\mathbf{r}) = |E_y(\mathbf{r})|^2, \quad (2)$$

which we use directly so that g reflects the true evanescent field at the trap height. Here $\omega_c = 2\pi f_{\text{res}}$, and g (together with κ and γ below) is an angular rate; we quote $g/2\pi$ in Hz. The dipole $\mu = 3.58 \times 10^{-29} \text{ C m}$ is the reduced matrix element $\langle J=\frac{1}{2} \| e\mathbf{r} \| J'=\frac{3}{2} \rangle$ of the ^{87}Rb D2 line; the reported g assumes the stretched σ^+ cycling transition (Clebsch–Gordan amplitude 1) and scales down by the corresponding amplitude for other transitions or polarizations (e.g. $\sqrt{3/5}$, $\sqrt{7/45}$, or the polarization-averaged $1/\sqrt{3}$), so it is not a universal Rb D2 value. The mode volume and Purcell factor are

$$V = \frac{\int \epsilon |\mathbf{E}|^2 dV}{\max(\epsilon |\mathbf{E}|^2)}, \quad F_P = \frac{3}{4\pi^2} \frac{Q}{V}, \quad (3)$$

with V expressed in $(\lambda/n)^3$. We normalize with the non-dispersive electric energy density $\epsilon |\mathbf{E}|^2$; replacing ϵ by the Brillouin form $\text{Re}[d(\omega\epsilon)/d\omega]$ for the dispersive SiN changes V by $\approx 1.5\%$, which we fold into the reported uncertainty. The total cavity energy-decay rate is $\kappa = \omega_c/Q$, and the cooperativity is

$$C = \frac{4g^2}{\kappa\gamma}, \quad (4)$$

where $\gamma = 2\pi \times 6.07 \text{ MHz}$ is the ^{87}Rb D2 natural linewidth. In the bad-cavity (Purcell) regime $\kappa \gg g \gg \gamma$ [20], an excited atom emits into the cavity mode rather than into free space with probability $C/(C+1)$. The fraction of cavity decay that exits through the intended collection waveguide is the branching ratio

$$\beta = \frac{\kappa_{\text{wg}}}{\kappa_{\text{tot}}}, \quad (5)$$

the cavity decay into the output ($-x$) waveguide normalized by the total. We report β with two estimators. The headline value is the directional-flux ratio $\beta_{\text{flux}} = \kappa(-x)/\sum_{\text{faces}} \kappa_{\text{dir}}$, the time-averaged Poynting flux through the output plane over the total directional flux. As a stricter cross-check, a frequency-domain mode-expansion monitor co-located with a flux monitor in the output waveguide projects the collection-plane field onto the precomputed guided modes, giving $\beta_{\text{guided}} = \beta_{\text{flux}} (P_{\text{guided}}/P_{\text{plane}})$ with $P_{\text{guided}} = \sum_m |a_m|^2$; since β_{flux} also counts radiation crossing the plane, $\beta_{\text{flux}} \geq$

β_{guided} . For the final device the two agree to within 0.3% and 99.5% of the collected power is in the fundamental guided mode, validating β_{flux} as a faithful branching ratio. Combining these, the efficiency with which the atomic excitation is delivered as a collected photon—the figure of merit we optimize—is

$$\eta = \beta \frac{C}{C+1} \eta_{\text{fb}} , \quad (6)$$

with $\eta_{\text{fb}} = 0.99$ an *assumed* external waveguide-to-fiber collection factor that we do not simulate. We therefore report both the chip-level efficiency $\eta_{\text{chip}} = \beta C/(C+1)$ and the full $\eta = \eta_{\text{chip}} \eta_{\text{fb}}$. Three distinct levers thus set η : the Purcell factor $C/(C+1)$ (emit into the cavity), the over-coupling β (route into the waveguide), and the collection η_{fb} . Crucially, because $C/(C+1)$ saturates once $C \gtrsim 10$, the relevant optimum maximizes η , *not* Q or C in isolation: pushing Q ever higher in a closed cavity drives $\beta \rightarrow 0$ and $\eta \rightarrow 0$ [21]. The residual infidelity of cavity-mediated entanglement scales as $\ln C/C$, so $C \sim 20$ already places the design in the regime of leading demonstrations. Finally, the photon is only emitted into the cavity if the resonance sits within roughly a linewidth of the atomic transition, $|f_{\text{res}} - f_{\text{D2}}| \lesssim \kappa/2\pi$; meeting this condition motivates the final trim.

C. Inverse design

We optimize the geometry with differentiable FDTD using the native adjoint capability of TIDY3D [15, 22]: a single forward frequency-domain simulation and one adjoint simulation yield the gradient of the objective with respect to every geometric degree of freedom. The cavity is parameterized by (i) a per-hole lattice constant a_i , so the hole placement schedule is free; (ii) per-hole in-plane scales $(s_{x,i}, s_{y,i})$; and (iii) a closed, free-form hole shape described by a periodic cubic B-spline of the radius as a function of polar angle. The radial (star-shaped) parameterization is closed and provably non-self-intersecting for any positive control radii, and x/y mirror symmetry reduces the twelve control points to four free radii per shape; at unit radius the hole reduces to an ellipse, so the seed is reproduced exactly. A fixed five-anchor blend (taper, mirror, and defect anchors) ties the per-hole shapes into a smooth schedule along the beam.

The loss combines the physics objective with fabrication-aware geometric barriers,

$$\mathcal{L} = -\log \tilde{\eta} + w_f \left(\frac{\hat{f} - f_{\text{D2}}}{\Delta f} \right)^2 + \mathcal{P}_{\text{geom}}, \quad (7)$$

where $\tilde{\eta}$ is a differentiable surrogate for Eq. (6) built from a softmax-peak-picked waveguide-to-total power ratio (for β) and a calibrated cooperativity proxy, $\hat{f}$ is the soft-resonance frequency, and $\mathcal{P}_{\text{geom}}$ collects fabrication-aware barriers enforcing a minimum dielectric gap, side-

rail width, spline curvature radius, and lattice smoothness (a gentle-confinement prior). Only frequency-domain field and mode monitors enter $\mathcal{L}$, since these are the differentiable quantities. We minimize $\mathcal{L}$ with Adam in two stages—first the lattice and hole-size layout, then the spline shapes—seeded by a Bloch band-structure design in which the mirror unit cell is sized so that its band gap is centered on the D2 line. Every candidate is independently verified outside the gradient loop by a time-domain FDTD simulation that extracts f_{res} and Q from the ring-down, the directional quality factors and hence β , the mode volume, and the field-sampled g , C , and η of Eqs. (2) and (4)–(6).

D. Resonance trim

A converged design generically lands tens of GHz from the atomic line. Because the in-plane geometry can be rescaled but the film thickness cannot, we tune the resonance with a single global conformal in-plane scale s applied to all hole positions, lattice constants, hole sizes, and spline vertices. Frequency scales approximately as $1/s$, but only approximately—the fixed thickness and the dispersive SiN index break exact scale invariance—so we measure $f_{\text{res}}(s)$ with a two-point bracket, interpolate linearly to the target, and apply one Newton refinement. Because a conformal scale preserves the *dimensionless* photonic design (the mirror gap and the cavity mode shift together), the trim changes β , C , and η by less than 1%.

E. Simulation parameters and convergence

Forward and verification simulations use the measured dispersive SiN index, an 8 ps run time (one to two cavity ring-downs, so the field-energy and flux monitors sample a strong-field epoch), perfectly matched layers on all boundaries, the y/z mirror symmetries of the E_y mode, and high-accuracy polarized subpixel averaging at the dielectric boundaries. We extract f_{res} and Q by two independent methods that agree to better than 10 MHz: a time-domain ring-down (resonance-finder) fit and a demodulated ring-down envelope fit.

The *absolute* resonant frequency requires care. On a single mesh it scatters by ~ 80 GHz (a few parts in 10^4) as the mesh is refined [Fig. 5(a)]—the signature of Yee-grid alignment noise at the curved hole boundaries of this high- Q ($Q \sim 10^4$) cavity. Staircased dielectric handling is far worse, while the choice of dispersive versus constant index at 780 nm is immaterial. We resolve this in three steps. (i) We refine the entire field-containing region (defect plus inner mirrors), not just the defect, with a uniform mesh down to 5 nm, so that the mirror Bragg condition that sets the resonance is fully resolved. (ii) We remove the residual grid-alignment noise by *grid-offset averaging*—averaging f_{res} over a set of sub-cell shifts of the structure at fixed mesh [Fig. 5(b)]; the offset-

averaged frequency is then mesh-consistent to 0.5 cavity linewidths between the 5 and 6 nm meshes. (iii) We re-derive the in-plane trim on this converged frequency. Because the fixed film thickness makes the in-plane scale only $\approx 60\%$ as effective as an ideal $1/s$ scaling, we calibrate the scale sensitivity from two measured points; the calibrated trim places the offset-averaged resonance at $|f_{\text{res}} - f_{\text{D2}}| = 0.3 \text{ GHz}$ with a numerical uncertainty of $\sim 23 \text{ GHz}$, i.e. definitively within one cavity linewidth ($\kappa/2\pi \approx 31 \text{ GHz}$). The branching ratio β , the efficiency η , and the mode profile are mesh-robust ($\lesssim 1\%$), and Q and f_{res} are stable (to $\lesssim 0.1\%$ and $< 5 \text{ GHz}$, respectively) across runtime ($8 \rightarrow 16 \text{ ps}$), PML thickness ($12 \rightarrow 18$ layers), and with the mirror symmetries disabled; the field-sampled g and hence C carry a larger $\sim 10\%$ mesh sensitivity, reflected in the uncertainties we quote.

III. RESULTS

A. Final device

Figure 1 shows the optimized cavity and its intended setting. The $25 \mu\text{m}$ -long nanobeam [Fig. 1(b)] comprises a short output coupling mirror, a graded defect that smoothly localizes the mode [Fig. 1(c)], and a long back reflector; the optimizer has shaped both the lattice schedule and the individual hole profiles. The released geometry exposes the mode to the trap region 250 nm above the surface [Fig. 1(d)].

B. Inverse-design trajectory

Figure 2 traces the optimization. The verified efficiency [Fig. 2(a)] climbs from $\eta = 0.01$ to 0.83 : the optimizer first opens the output port—rapidly raising β and hence η —reaching near-final efficiency by the end of the layout stage, after which the shape stage refines the mode. The corresponding verified loss $-\log \eta$ [Fig. 2(b)] falls monotonically (best-so-far) from 4.6 to 0.18 , confirming that the delivered design is the best the optimizer found; the full proxy objective additionally carries a resonance-penalty term whose transients are absorbed by the final trim. The structure snapshots [Fig. 2(c)–(e)] show the hole schedule evolving from the band-engineered seed to the final graded profile.

C. Mode structure and band engineering

The verified mode [Fig. 3(a),(c)–(e)] is a single, tightly localized state centered on the defect, with an evanescent tail that reaches the 250 nm trap height. Its photonic origin is shown in Fig. 3(b): the mirror unit cell exhibits a transverse-electric photonic band gap [23] that brackets the D2 frequency, providing in-plane confinement, while the graded defect raises a band edge to the D2 line to

localize the mode. This is the band-engineering that the seed supplies and that the inverse design preserves.

D. Performance and design lineage

The final trimmed device sits on the ^{87}Rb D2 line: a grid-offset-averaged, mesh-converged analysis places the resonance at $|f_{\text{res}} - f_{\text{D2}}| = 0.3 \text{ GHz}$ with a $\sim 23 \text{ GHz}$ numerical uncertainty—within one cavity linewidth. Its loaded quality factor is $Q = 1.2 \times 10^4$ ($\kappa/2\pi = 31 \text{ GHz}$), with a guided-mode branching ratio $\beta = 0.88$ (the directional-flux and mode-expansion estimators agree to 0.3% , and 99.5% of the collected power is in the fundamental waveguide mode), cooperativity $C \approx 22$, single-photon coupling $g/2\pi \approx 1.0 \text{ GHz}$ at the 250 nm trap height, mode volume $V = 0.89 (\lambda/n)^3$, and Purcell factor $F_P \approx 1.0 \times 10^3$. The chip-level interface efficiency is $\eta_{\text{chip}} = 0.84$ ($\eta = 0.83$ including the assumed fiber factor). The loss budget is dominated by the intended channel: $\approx 88\%$ of the cavity decay exits the output waveguide, with only $\sim 12\%$ residual scattering, confirming the device is strongly over-coupled rather than merely high- Q .

Figure 4 places this result in context. Panels (a)–(d) show the verified metrics climbing from the seed: the optimizer raises η and β by an order of magnitude past the hand-tuned baseline (dashed), while the cooperativity settles near $C \sim 22$ and the bare Q is deliberately *reduced* from the seed’s value—the design trades useless intrinsic Q for a useful, over-coupled interface. The design lineage [Fig. 4(e),(f)] makes the same point across four geometries: the hand-tuned baseline and the band-engineered seed are both strongly under-coupled ($\beta \sim 0.09$, $\eta \sim 0.08$), even though the seed achieves a nominally enormous intrinsic quality factor; gradient inverse design opens the output port to $\beta = 0.88$ and $\eta = 0.83$, and the conformal trim then lands the resonance on the atomic line [Fig. 4(g)] without disturbing that interface. Panel (h) shows the over-coupling trade-off directly: as the output mirror is made more transparent (the external coupling κ_{wg} increased), the loaded Q and the cooperativity C fall while β rises toward unity; because η [Eq. (6)] multiplies a rising β against a falling cavity-emission factor $C/(C+1)$, the optimum sits at intermediate, strong-but-not-maximal coupling, coinciding with neither maximal Q nor maximal C . We caution that the band-engineered seed’s apparent $Q \sim 10^7$ and $C \sim 10^5$ are unresolved upper bounds—the field barely decays within the simulation window—and not operating values; the operating design is the resolved- Q , over-coupled trimmed cavity.

Relative to the hand-tuned baseline, the final design improves the transduction efficiency by roughly a factor of ten ($\eta : 0.083 \rightarrow 0.83$) and clears both the $\beta \geq 0.80$ and $\eta \geq 0.75$ targets. It is single-mode—the nearest competing resonance lies more than 1 THz away—and the bad-cavity hierarchy $\kappa \gg g \gg \gamma$ is maintained. Across

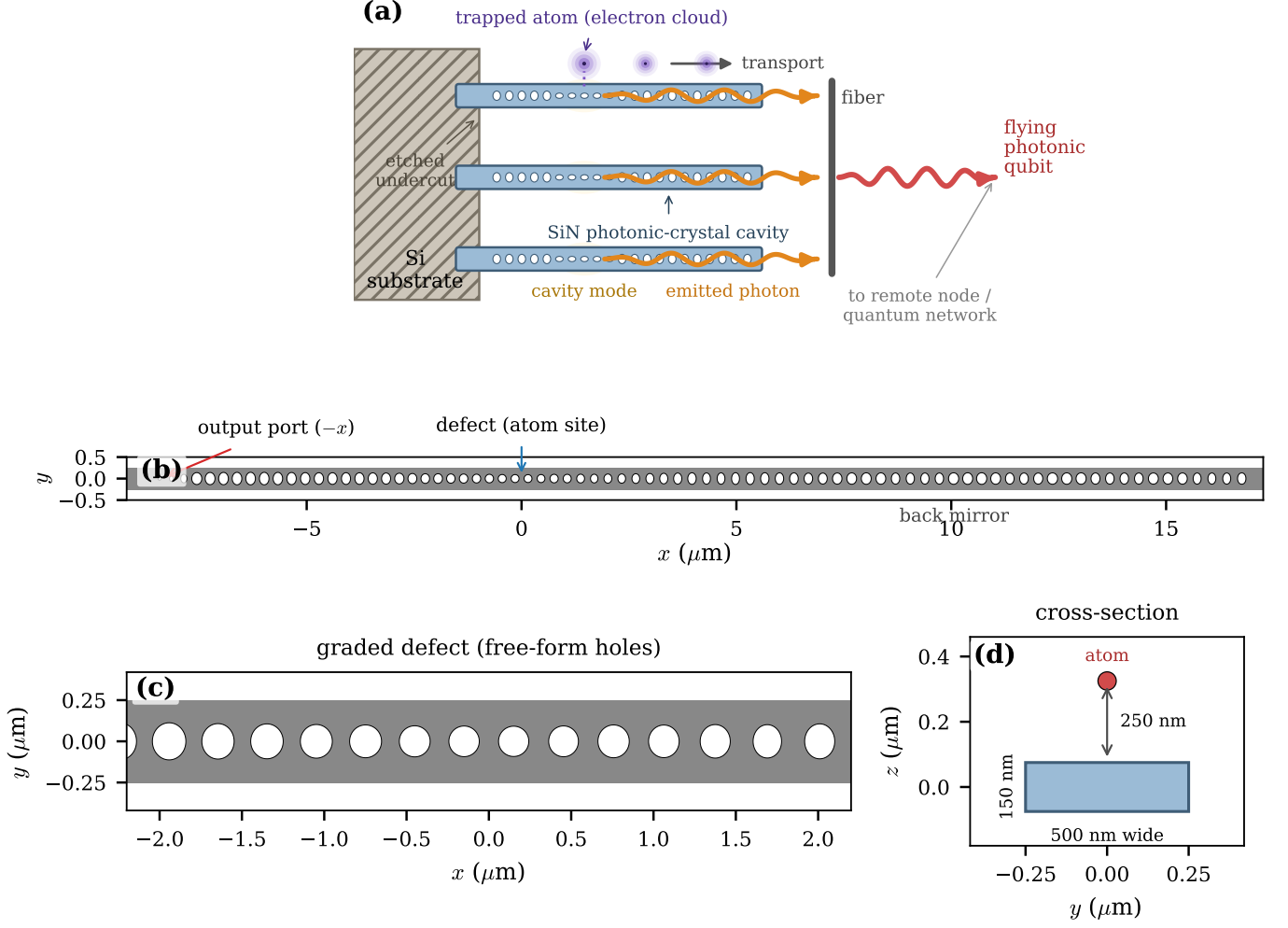


FIG. 1. **Platform and final device.** (a) Concept: an array of suspended SiN photonic-crystal “fingers” cantilevered beyond an etched substrate edge; tweezer-trapped atoms are transported over the series of cavities and transduced into flying photonic qubits collected into fiber. (b) Top view of the inverse-designed cavity, with the asymmetric output ($-x$) coupling mirror, the central defect (atom site), and the long back mirror. (c) Zoom of the graded defect region showing the free-form spline holes and the gentle lattice taper. (d) Beam cross-section: the 500 nm-wide, 150 nm-thick nanobeam with the atom trapped 250 nm above the top surface.

a deterministic fabrication corner grid (hole-radius etch bias ± 4 nm, lattice constant ± 2 nm, beam width ± 10 nm) the efficiency stays in the range $\eta = 0.82$ – 0.86 and the branching ratio $\beta = 0.86$ – 0.89 : the perturbations shift the absolute resonance, which the same global-scale trim restores, but leave the interface metrics essentially intact.

E. Likely operational impact

Interpreted strictly as electromagnetic proxies, these numbers indicate a substantial improvement for the atom-to-flying-qubit channel. The cooperativity rises from $C \approx 0.07$ in the first single-atom photonic-crystal-cavity coupling [12] to $C \approx 22$ here, placing the cavity-emission efficiency $C/(C+1)$ near unity, and the mode-expansion-validated over-coupling $\beta = 0.88$ delivers the

photon into a single waveguide mode. Taken purely at face value, a two-node heralding rate scaling as η^2 would then improve by roughly two orders of magnitude over that early demonstration; we stress that this is an arithmetic extrapolation from the proxies—it ignores detector efficiency, mode overlap, phase stability, and atom survival—and that the often-quoted cavity-protocol infidelity scaling $\sim \ln C/C$ (≈ 0.14 at $C \approx 22$) is a heuristic we do not evaluate here. All such statements are projections from the classical interface metrics, not results of a quantum-dynamics simulation.

IV. DISCUSSION

We have inverse-designed a suspended-SiN photonic-crystal nanobeam cavity, resonant with the ^{87}Rb D2 line,

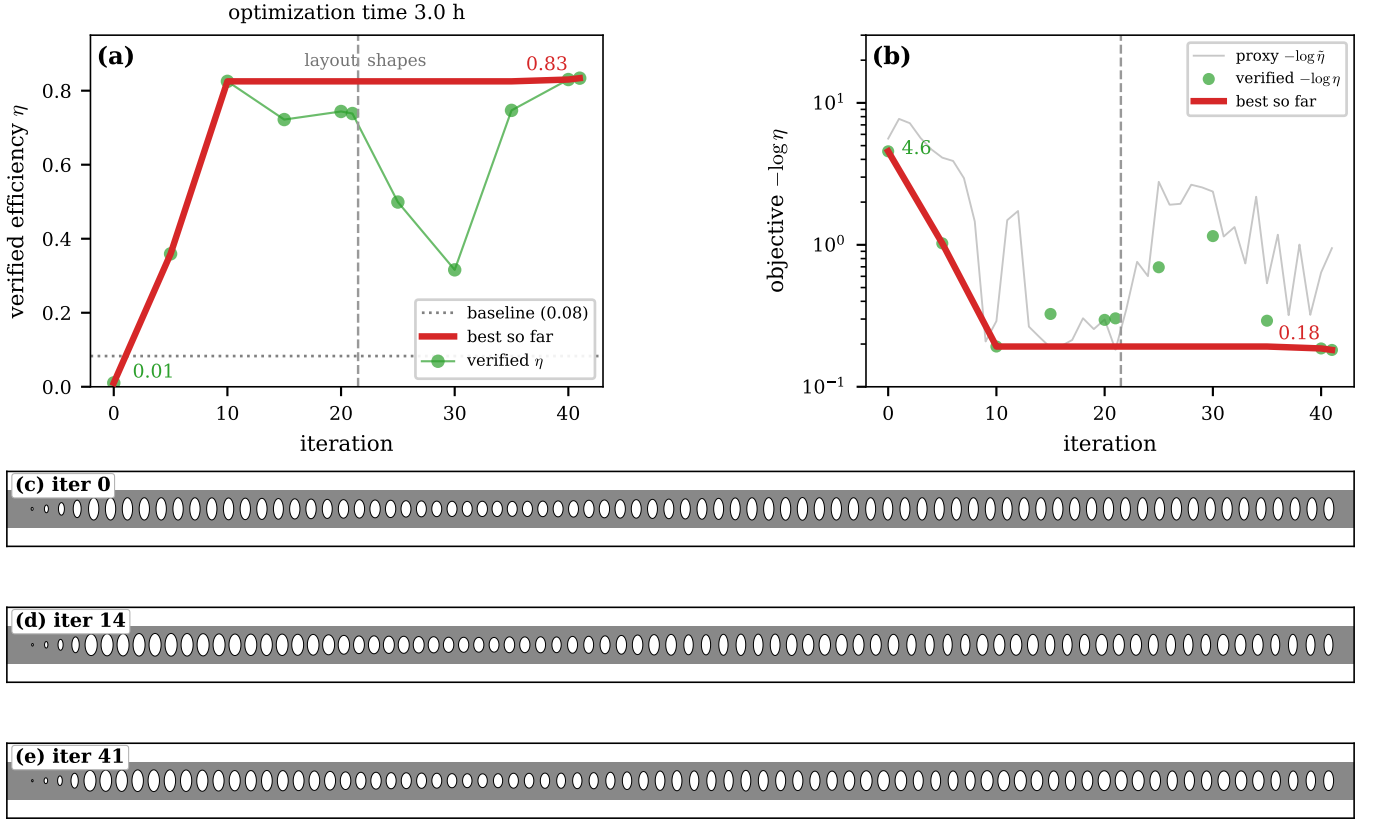


FIG. 2. **Inverse-design trajectory.** (a) Verified interface efficiency η from the periodic non-gradient (high-resolution time-domain) checks versus iteration: it climbs from 0.01 to 0.83, and the best-so-far envelope (red) shows that the optimizer reaches its final efficiency by the end of the layout stage, with the shape stage refining it. The dashed line marks the layout→shapes boundary; the dotted line is the hand-tuned baseline. (b) Optimization objective versus iteration: the verified interface loss $-\log \eta$ from the non-gradient checks falls from 4.6 to 0.18, and its best-so-far envelope decreases monotonically to the delivered design. The faint trace is the proxy surrogate $-\log \tilde{\eta}$ actually minimized at each iteration. The full proxy objective [Eq. (7)] additionally carries a resonance-penalty term whose transients are absorbed by the final trim and are not shown. (c)–(e) Top-view snapshots of the structure at early, intermediate, and final iterations, showing the evolving hole schedule.

that directly optimizes the atom-to-waveguide transduction efficiency. The approach factorizes cleanly: Bloch band engineering supplies a clean photonic foundation with the mirror gap on the atomic line, while adjoint inverse design supplies the over-coupled, on-resonance interface; a conformal in-plane scale lands the mode on the line without disturbing the dimensionless design. The result, $\eta_{\text{chip}} = 0.84$ with a guided-mode $\beta = 0.88$ and $C \approx 22$, is a tenfold efficiency gain over a hand-tuned baseline and a two-order-of-magnitude cooperativity gain over the first single-atom photonic-crystal-cavity demonstration.

Several limitations bound these claims, and we state them quantitatively. (i) g , C , β , and η are fixed-position, two-level, dipole-aligned scalar proxies extracted from classical fields, not outputs of a master-equation or quantum-trajectory simulation; we make no claim about gate, readout, or entanglement fidelities, atom motion, or multilevel dynamics. (ii) The reported g assumes the σ^+ cycling transition; other transitions or polarizations scale it down by their Clebsch–Gordan amplitude. (iii) The *ab-*

solute resonance carries a ~ 23 GHz (sub-linewidth) numerical uncertainty even after grid-offset-averaged mesh convergence, and the field-sampled g and hence C carry a $\sim 10\%$ mesh sensitivity, while β and η are mesh-robust to $\lesssim 1\%$. (iv) β is a directional-flux ratio cross-checked by mode expansion (the two agree to 0.3%), not a measured quantity. (v) The band-engineered seed’s high Q and C are unresolved upper bounds, not operating values, and the atom-position maps hold κ and β fixed at the cavity values. (vi) The device has not been fabricated; the fabrication corner grid reports interface metrics under an ideal global-scale retrim, so the untrimmed resonance yield—which the in-plane trim, a pre-fabrication design knob, would need to recover—is a separate question, and the trim is in-plane only, fixed by the deposited film thickness.

These limitations point to natural next steps. A full cavity-QED treatment—a master-equation model including the multilevel atom, motion, and realistic decoherence—would convert the present interface proxies into protocol-level fidelity and rate predictions. Fab-

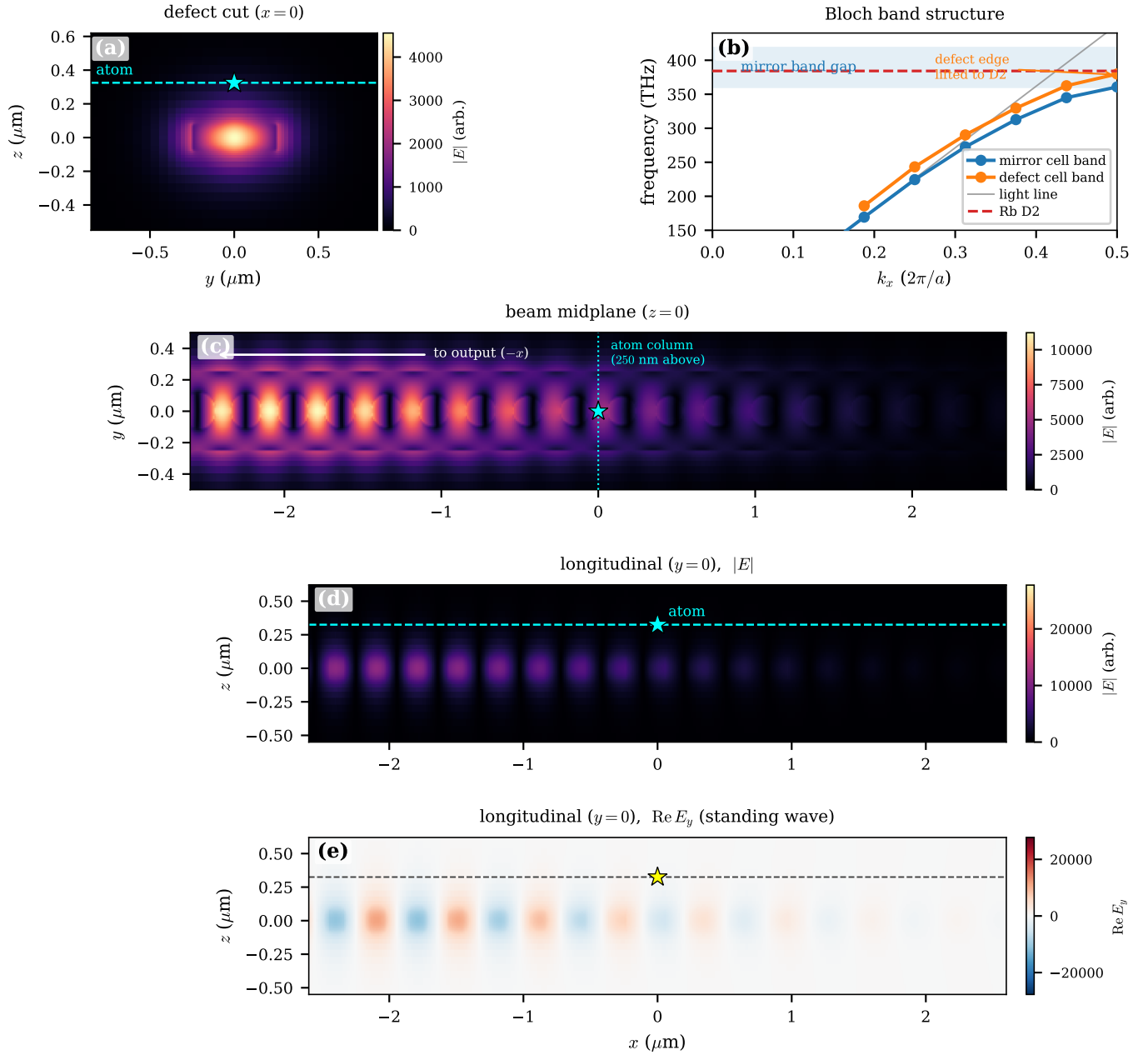


FIG. 3. **Cavity mode and band structure.** (a) Electric-field magnitude $|E|$ in the defect cross-section ($x=0$); the star, on the dashed line, marks the trapped atom 250 nm above the top surface, where the evanescent field couples to it. (b) TE Bloch band structure: for each unit cell the lowest guided band is traced (markers, from the unit-cell ring-down). The mirror band edge opens a photonic band gap (shading) [23] that brackets the Rb D2 line (dashed), giving in-plane confinement; the defect cell's band edge is lifted into this gap to the D2 line, localizing the mode (light line shown for reference). (c) $|E|$ in the beam midplane ($z=0$); the atom sits on the marked column at $x=0$, 250 nm above this plane. The standing-wave envelope leans toward the $-x$ output because the cavity is strongly over-coupled to that port ($\beta=0.88$), extending preferentially toward it rather than peaking at the defect. (d) $|E|$ and (e) $\text{Re } E_y$ in the longitudinal plane ($y=0$, atom marked); (e) resolves the standing-wave sign structure of the confined mode. Panels (c)–(e) share the true-aspect x axis.

rication and cold-atom characterization would test the design against real disorder and the trap geometry; post-fabrication thermal or MEMS tuning would add in-situ resonance control to complement the design-stage trim demonstrated here. Finally, the single-cavity design is

the unit cell of the finger-array architecture of Fig. 1(a): extending the optimization to multi-finger arrays with shared collection, and allowing a second adjoint stage to push β further, are the path toward a scalable atom-array photonic interconnect.

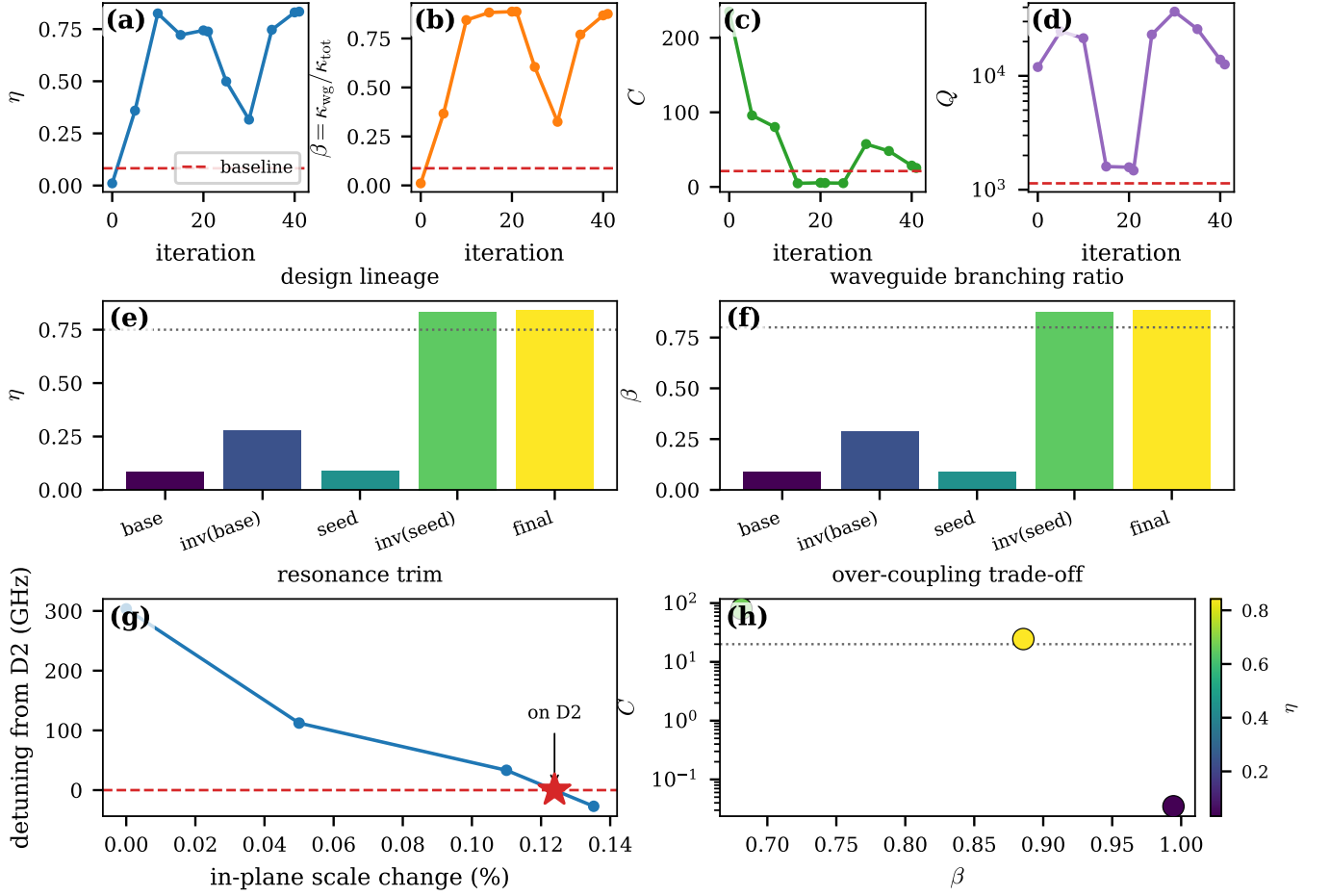


FIG. 4. **Performance evolution, lineage, and tuning.** (a)–(d) Verified η , β , C , and Q versus iteration; dashed lines mark the hand-tuned baseline. The optimizer raises η and β by an order of magnitude while trading the seed’s excess intrinsic Q for a useful, over-coupled interface. (e), (f) Efficiency η and branching ratio β across the design lineage (baseline; gradient-only; band-engineered seed; seeded inverse design; final trimmed); dotted lines mark the 0.75 and 0.80 targets. (g) Resonance trim: detuning from the D2 line versus a global in-plane scale; the star marks the selected on-resonance design. (h) Over-coupling trade-off: cooperativity C versus branching ratio β (color: η) as the output mirror is made progressively more transparent (the external coupling κ_{wg} increasing); raising κ_{wg} lowers C and the loaded Q while raising β , so maximal η lies at intermediate coupling, not at maximal C or Q .

RESOURCES

Hardware. Electromagnetic simulations were run on the TIDY3D cloud finite-difference time-domain solver (GPU-accelerated, billed in FlexCredits); geometry construction, optimization bookkeeping, and postprocessing ran on a local workstation.

Software. Differentiable FDTD and band-structure simulations used TIDY3D 2.11.2 with its native automatic-differentiation (adjoint) capability [22]; gradients were propagated with AUTOGRA and the optimization used Adam. Geometry, postprocessing, and figures used NUMPY, SCIPY, SHAPELY (polygon validity), GDS-FACTORY (GDS export), and MATPLOTLIB. Bloch band

structures were computed with TIDY3D unit-cell simulations.

Data and code availability. The full pipeline is available: the differentiable inverse-design and optimization code; an auditable high-accuracy validation script (`validate_design.py`) that recomputes every figure of merit with units shown, runs the mesh and grid-offset frequency-convergence study, and extracts the mode-expansion branching ratio; the verified per-design datasets (HDF5 monitor data and JSON/CSV figure-of-merit tables with an embedded metric-convention block) underlying every figure; the final-device geometry (GDS and per-hole parameter tables); and one-command regeneration scripts. A repository DOI will be provided on publication.

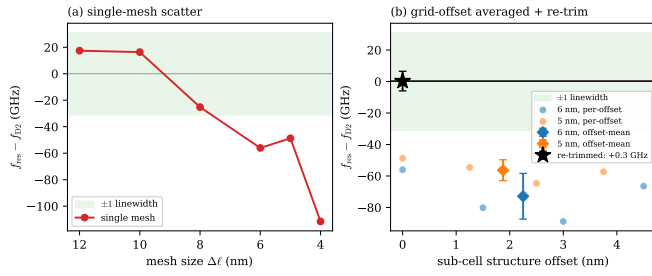


FIG. 5. **Resonant-frequency convergence.** (a) On a single mesh, the absolute f_{res} scatters by ~ 80 – 130 GHz as the mesh is refined—grid-alignment noise at the curved hole boundaries (shading: one cavity linewidth). (b) Grid-offset averaging (averaging f_{res} over sub-cell shifts of the structure at fixed mesh) collapses the noise: the offset-mean (diamonds) is mesh-consistent to 0.5 linewidths between 5 and 6 nm, and the sensitivity-calibrated in-plane trim places the resonance on the D2 line (star, $+0.3$ GHz), within one linewidth.

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